

Digital Material Passport

ID TEST-SPECIMEN-001 - Version 0.1.0
Issue Date 2025-09-29 - Certificate Type

Manufacturer	Customer	Business Transaction
Steel Testing Labs GmbH 123 Test Street 10115 Berlin DE	Customer Industries Inc 456 Customer Ave New York 12345 US	Order ID ORD-2025-001 Delivery ID DEL-2025-001 - Quantity 1000 kg

Product

Product Name	High Strength Steel Plate	Shape	Plate - Width 2000 mm - Thickness 50 mm - Length 6000 mm
Batch ID	BATCH-2025-001		

Chemical Analysis

Heat Number **H123456** - Melting Process **EAF+LF** - Casting Method **ContinuousCasting** - Casting Date **2025-09-28**

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
C	%		0.2	0.18	Ni	%		0.2	0.15
Mn	%	1.4	1.7	1.45	Mo	%		0.1	0.08
Si	%		0.5	0.35	Cu	%		0.25	0.18
P	%		0.025	0.012	Al	%	0.02	0.05	0.025
S	%		0.015	0.008	N	%		0.012	0.008
Cr	%		0.3	0.25	CEV	%		0.45	0.42

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength ¹ Specimen: 1/4T, L	Rm	485 MPa	450 MPa	600 MPa	ASTM E8	✓
Charpy V-Notch Impact Temperature: -40°C Specimen: 1/4T, L-T	KV	48 / 52 / 45 J - Average 48.3 - Min 45 - Max 52 - Std Dev 3.5	40 J		ASTM E23	✓
Through-Thickness Tensile Room temperature, through-thickness direction					EN 10164	✓

Location (position)	0	0.25	0.5	0.75	1
Value [MPa]	475	485	490	487	478

Yield Strength ¹ Specimen: Surface, T - ID: YS-SURF-T-001	Re	355 MPa	335 MPa	ASTM E8	✓
Elongation ¹ Specimen: Custom (Mid-radius at end section), C - ID: ELONG-MR-001	A	22 %	20 %	ASTM E8	✓

¹ Room temperature

Validation

We certify that the material described herein has been tested and inspected in accordance with the specified standards and meets all requirements.

Validated by **Dr. Hans Schmidt**, Quality Manager - 2025-09-29